

GGPL ENQUIRY BOT

COMPLETE TRAINING GUIDE

Customer Description to GGPL Quote Description - Full Rule System
Core Conversion Rules + Hostile Input Handling (Rules B-S) + Audit Mode

Built from	35,000+ GGPL historical conversion pairs, material masters, FEEDBACK corrections, and 20 real-world 'monster format' case studies
Verified against	ASME B16.20-2017, B16.21-2021, B16.47, API 6A / ISO 10423, DIN 2696, EN 1514
Architecture	Two-stage: LLM extracts and classifies to JSON; deterministic code assembles the GGPL string via templates and lookup tables
Applies to	Both Excel attachments AND copy-paste input (ingestion equivalence, Rule R)
Version	V6 Consolidated - July 2026

How to use this guide to teach the model: Sections 1-3 become the system prompt (classification + extraction + hostile-input handling). Sections 4-6 (templates, lookup tables, dictionaries) are implemented as deterministic code and editable data tables, NOT prompt text. Section 8 fixtures become the permanent regression suite - every prompt or model change must pass them before release.

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4	Product Templates Quick Reference	SC, SPW, RTJ, KAMM, DJI, ISK, specials
5	Lookup Tables	Ring numbers, hardness, conversions, DN/NB map
6	Dictionaries	Materials, fillers, brands (3 buckets), abbreviations, synonyms
7	Escalations, Validation and Audit Mode	Exact phrases, 20-point checklist, audit workflow
8	Regression Fixtures	The monster test set the bot must always pass

SECTION 1 - THE PIPELINE AND GOLDEN PRINCIPLES

1.1 Two-stage architecture (why the bot makes no formatting mistakes)

The LLM's ONLY job is to read hostile input and produce structured JSON: product class + extracted fields + flags. Deterministic code then assembles the GGPL string using templates and lookup tables. Code never mistypes a standard, never swaps a filler, never invents a ring number. Every rule in this guide states which stage owns it.

```
RAW INPUT (Excel / paste / email / PDF text)
| Stage 0: NORMALIZE - same intermediate table for every input path
| Stage 1: LLM       - pre-parse, classify per row, extract JSON, raise flags
| Stage 2: CODE      - size world -> template -> lookups -> assemble string
| Stage 3: VALIDATE  - 20-point checklist (Sec 7.2); fail -> human queue
v Stage 4: RECONCILE - "N lines parsed, qty column used, X escalations"
GGPL QUOTE LINES + side-fields (refs, qty, certs, flags, deviations)
```

1.2 The three size worlds - detect FIRST, decides everything

World	Signals	Output size format	Standards family
W1 ASME inch/class	NPS, inch marks, #, LB, CL, Class 150-2500	{x}" X {ppp}#	ASME B16.21 / B16.20 / B16.47 / API 6A
W2 Metric DN/NB/PN	DN, NB, PN, EN/DIN/ISO/BS10 refs	{x} DN X PN{x} or NB {x} X {ppp}# - NEVER an inch mark on DN/NB	EN 1514, EN 1092, ISO 4633, EN 681, BS 10
W3 Non-standard dims	OD/ID/LW in mm, drawing refs, equipment gaskets	{OD}MM OD X {ID}MM ID X {t}MM THK - OD first, OD > ID	none / (AS PER DRAWING) / (NON STANDARD)

PN present means the size is DN even if unlabeled (25PN16 -> DN 25). Size over 3 digits with PN is W2/W3, never inches. NEVER convert between worlds - quote in the customer's world. mm pipe-OD in a W1 flange context maps to NPS: 21.3->1/2", 26.7->3/4", 33.4->1", 48.3->1-1/2", 60.3->2", 73/76.1->2-1/2", 88.9->3", 114.3->4", 168.3->6", 219.1->8", 273->10", 323.9->12".

1.3 The six fidelity laws (violations = severe errors)

#	Law	Meaning
1	Filler fidelity	Stated filler copied EXACTLY (CNAF, PTFE, GRAPH 98%, VERMICULITE grade). Graphite is only the silence-default. Historic bot failure: swapped stated CNAF for graphite.
2	Material fidelity	Normalize the name, never change the alloy. Grade suffixes survive: 304H is not 304; 316L is not 316; 347 is not 316L (real audit catch).
3	Both-rings law	IR and OR both stated means both appear in output. Historic failure: missed OR.
4	Shape fidelity	OVAL stays OVAL. RX/BX are ring TYPES, never shapes.
5	Verbatim brands	Product-identity brands (Kroll & Ziller) copied exactly - house rule.
6	No guessed construction	Rib, fire-safety, ring number, ID-vs-OD: derived from rules or asked. Never invented.

1.4 Output grammar

UPPERCASE. Prefix SIZE: . Size elements joined by X . Comma-separated fields. Thickness as {t}MM THK (decimal point). Coatings retained (GALVANISED / ZINC / CADMIUM PLATED / ELECTROPLATED / XYLAN / PTFE / EPOXY COATED). Drawing reference anywhere -> append (AS PER DRAWING). NACE: dropped by default, EXCEPT the RTJ RX template (always includes it) and explicit certification demand (append NACE MR0175, before the standard). Canonical spellings: OCTAGONAL, INCOLOY, INCONEL, HASTELLOYS, GALVANISED, SILICONE, KAMMPROFILE, ASME B16.20.

Never in the GGPL string (side-fields only): tag/PL/cat/item codes, 'for B16.5 flanges', service notes, Lethal, oil quality, QA/QC cert lines, O-ring bar ratings, pipe schedules, quantities, project names, AVL mentions.

SECTION 2 - CLASSIFICATION AND STANDARDS ENGINE

2.1 Classification priority (first match wins - run PER ROW, never per table)

1 ISK -> 2 RTJ -> 3 LENS -> 4 O-RING/CORD -> 5 DJI ->
6 KAMM (incl. "PROFILE GASKET") -> 7 SPW (incl. MC/MCR manhole spiral) ->
8 CORRUGATED METAL (CMG) -> 9 MANHOLE/HANDHOLE -> 10 ENVELOPE ->
11 METAL CLAD -> 12 SOLID METAL FLAT -> 13 PLUG -> 14 LIP SEAL ->
15 DIAPHRAGM -> 16 EYELET -> 17 SHEET/ROLL -> 18 SC (catch-all flats) ->
19 ADJACENT/REGRET list. Unclear -> KINDLY PROVIDE CLEAR SPEC

Critical traps: 'PROFILE GASKET' is ALWAYS KAMM, even with rubber/FKM facing (bot's most famous historical misclassification). 'Corrugated type ... filler' INSIDE a DJ description stays DJI; standalone 'corrugated gasket' is CMG. 'Lip seal' as a noun is a product; 'WITH LIP' as a shape stays in the parent family. 'O-RING' spelling variants: "O" RING / O RING / ORING / O.RING.

2.2 Standards engine - computed from (product, size, class), NEVER copied from customer text

Product	<= 24 inch	26-60 inch	W2 metric	API
SC	ASME B16.21	ASME B16.47 (SERIES A)	EN 1514-1 / customer std	-
SPW	ASME B16.20	B16.47 (SERIES A; B if stated)	B16.20 / EN 1514-2	NEVER API
KAMM	ASME B16.20	B16.47 (A/B as stated)	EN 1514-6	-
DJI	ASME B16.20	B16.47	-	-
RTJ	ASME B16.20 (R rings)	B16.47 (SERIES-A) / R93-R105	-	API 6A (BX), API 6B (RX)
ISK	to suit B16.5	to suit B16.47 (SERIES-A)	to suit EN 1092	to suit API 6A
ENVELOPE	B16.21	B16.47 (A)	EN 1514-3	-
LENS	-	-	DIN 2696	-

Class gaps (nonexistent flanges - flag, never force): no CL400 at NPS 1/2-3 (use 600); no CL900 at NPS 1/2-2-1/2 (use 1500 dims); no CL2500 at NPS 14+. 16 inch 2500# does NOT exist in B16.5 - a client special flange spec (SP-xxx code) -> dimensions escalation. **Obsolete standards dictionary:** API 601 -> ASME B16.20; API 605 -> ASME B16.47 (SERIES B); MSS SP-44 -> B16.47 (SERIES A); ANSI -> ASME (same number). Quote the successor + deviation note.

2.3 Defaults (only when customer is silent - see exhaustive-list exception, Rule F)

Field	Default
SPW thickness	4.5MM THK (the 4.45mm nominal). Soft Cut 3MM. KAMM std flange 4.5MM total.
SPW outer ring	CS. Inner ring = winding material. Shorthand 'I & O RING-SS304' = both SS304.
SPW winding absent, rings stated	Winding = ring material, logged as assumption (SAP key-value pattern).
Inner ring MANDATE (B16.20)	CL900 NPS 24+; CL1500 NPS 12+; CL2500 NPS 4+; ALL PTFE-filled. Customer omitted it? ADD it + note. Mandate beats even the exhaustive-list rule.
RTJ shape	OCTAGONAL (keep OVAL if stated). Soft Cut facing: ring type RF, full face FF.
DJI / KAMM rib	NEVER guessed - quote + KINDLY CONFIRM RIB DETAILS.
ISK fire status	Always print one: MICA/secondary seal or fire-tested -> (FIRE SAFE); PTFE-only / phenolic / no core -> (NON FIRE SAFE).

SECTION 3 - HOSTILE INPUT HANDLING: RULES B TO S

Each rule was reverse-engineered from a real customer file that confused earlier bot versions. Format: DETECT (when it fires) / DO (the behaviour) / EXAMPLE (from the actual monster).

RULE B - STRUCTURED KEY-VALUE ENQUIRIES (SAP characteristics / datasheet format)

DETECT: 3+ lines of 'Label : Value' or a two-column Characteristics/Value table (Gasket Type, Size, Part Size Units, ANSI Class, Note/Note-n, Manufacturer, QSCert/QACert, Base Unit of Measure...).

- Decode entities FIRST: & -> &, <(>&<)> (SAP escape) -> &, " -> quote.
- PIVOT all rows of a block into ONE line item. Pair unit fields: Size 3/4 + Units INCH -> 3/4".
- Run every Note-n through the free-text extractor - **notes outrank labeled fields** on conflict (notes carry the engineer's intent).
- Winding inference: rings stated + winding absent -> winding = ring material + assumption flag.
- Cert fields (IGC, PMI, MTC 3.1/3.2, conformity, origin) -> requirements side-channel, never the string. EXCEPTION: NACE = Yes is an explicit demand -> 'NACE MR0175,' before the standard.
- MANUFACTURER field holding an END USER / AVL name (ADNOC, ARAMCO, PDO, QP, KOC, Shell) = commercial flag 'AVL approval required', NOT a gasket brand. A gasket maker name -> brand dictionary.
- Abbreviation lexicon: FRSFE/FRSAFE->FIRE SAFE, INSLTN->INSULATION, SRVC->SERVICE, COATD->COATED, DI-ELECTRIC->DIELECTRIC. Field value '---' = per-mfr-std / not in scope - NEVER an escalation trigger.

```
IN : Gasket Type: Spiral Wound | Inner ring material: 316/316L | Gasket Size: 3/4
    Part Size Units: INCH | ANSI Class: 600 | Note: INNER & OUTER MATERIAL SS316/L
    Note2: FILLER MATERIAL GRAPHITE | Manufacturer: ADNOC AVL | QACert-NACE MR 0-175: Yes
OUT: SIZE: 3/4" X 600# X 4.5MM THK,SS316/SS316L SPIRAL WOUND GASKET WITH GRAPHITE
    FILLER + SS316/SS316L INNER RING & SS316/SS316L OUTER RING,NACE MR0175,ASME B16.20
FLAGS: ADNOC AVL APPROVAL REQUIRED | IGC+PMI+MTC 3.1 -> requirements | winding=rings (assumed)
```

RULE C - MULTI-ROW PO TABLES + THE CONFLICT HIERARCHY

DETECT: Header row with 2+ of {Material, Item Code, Description, Short Text, Qty, PO text, Long Text, UOM, Sr.No, Unit, PCE/EA/NOS/SET}; rows anchored by a material-code pattern.

- One output line per material-code row; Qty + code -> side fields. Rejoin line-wraps inside cells BEFORE parsing (ASME B / 16.5 split across lines is one token).
- **CONFLICT HIERARCHY: standards engine > PO long text > short description.** A short text citing B16.21 on a spiral wound item is ERP master-data residue - SPW is B16.20 regardless. Log every override in deviation; never argue it in the GGPL string.
- Compact tokens: '{METAL}/{FILLER}' pairs (SS 316/FLX GRPH) -> winding + filler. FLX GRPH/FG/GRPH -> graphite family. Quality Requirements / MTC lines -> side-channel.
- FAMILY VALIDATION with the COHERENCE TEST: single-field drift across a copy-paste family (one row's filler differs, FLG STD ticks up) = suspected typo -> flag. MULTI-field coherent deviation (Type E->D together with FF->RTJ together with a class jump) = genuine variant -> quote as written, no flag. Test: do the changed fields make engineering sense TOGETHER?

```
IN : short text: GSKT SPW 1 1/2" B16.21 #150 SS316 4.5THK
    PO text: SIZE 1 1/2"; RATING 150#; SP.WINDING 4.5MM SS 316/FLX GRPH; STD B16.20/B16.5
OUT: SIZE: 1-1/2" X 150# X 4.5MM THK,SS316 SPIRAL WOUND GASKET WITH FLEXIBLE GRAPHITE
    FILLER + SS316 INNER RING & CS OUTER RING,ASME B16.20
DEV: SHORT TEXT CITES B16.21 - SPW QUOTED TO ASME B16.20 (CORRECT STANDARD)
```

RULE D - MTO / BOQ EMAILS + THE AUTO-INCREMENT ARTIFACT

DETECT: Group headers (equipment tags ME-/E-/HE-xxx, MTO/BOQ numbers) with item rows beneath; row labels like Gasket(#150) 4; PDF/email artifacts present.

- SCRUB before parsing: page headers ('2 of 3'), email headers, signatures. Rejoin text severed by page breaks (spec interrupted mid-word resumes after the artifact).
- Structure: equipment -> MTO group -> items; carry both refs per line. Empty groups skipped.

- Row grammar: Gasket({n}{ppp}) {size}; decimals are inches (0.5=1/2", 0.75=3/4", 1.5=1-1/2"). Lone trailing integer = qty. Also '{n}mm{WORD}' and '{n}CL' glued sizes -> split.
- **AUTO-INCREMENT DETECTION:** a field ticking up by exactly one step across consecutive rows while all else is identical (FLG STD B16.5 -> B16.6 -> ... -> B16.12) is an Excel fill-handle artifact. Corroborate: B16.9 = buttweld fittings, B16.10 = valves - not flange standards. Normalize ALL rows to the family base value + ONE deviation note.
- RECONCILE: always report 'N items parsed: breakdown' back to the user - instant trust check.

```
IN : 19 rows FLG STD B16.5, then B16.6, B16.7 ... B16.12 on sequential rows (drag-fill)
OUT: all rows -> ASME B16.5 flanges, gaskets to ASME B16.20
DEV: FLG STD B16.6-B16.12 ON SEQUENTIAL ROWS - EXCEL AUTO-INCREMENT ARTIFACT, ALL TAKEN AS B16.5
CHECK: 32 items parsed: 19 ME-102 + 13 ME-301
```

RULE E - COLUMN-ROLE VALIDATION + PER-ROW CLASSIFICATION

DETECT: Any table where headers might lie: class values sitting in a Sch column, generic 'Gasket' description columns, spec hidden in MOC/Remark columns.

- HEADERS ARE HINTS, NOT TRUTH. Validate each cell against content patterns: {n}# / CL{n} / {n} LB = CLASS wherever it sits; {n}" / DN{n} = SIZE; SCH 40/80/XS = pipe schedule -> side-field.
- Generic description columns ('Gasket', 'Seal') carry no signal - classify from the spec column.
- **CLASSIFY EVERY ROW INDEPENDENTLY.** One table is not one product: an 'Oval ring Type R' row is RTJ and a 'profile gasket' row is KAMM even sandwiched between spiral wound rows. Inherit from the previous row ONLY on explicit ditto marks (-do-, ").
- Qualifiers are spec: filler purity (98% pure), 'Type R', shape words - carried into output.
- 'Thickness as per manufacturing std' = apply GGPL default + log assumption (invitation, not omission).
- RFQ preamble mining: project name, end user, AVL, 'acknowledge receipt', bid dates -> commercial flags.

```
IN : columns Sr|Desc|Size|Sch|Class|MOC ... 300#/600#/1500# sit in Sch, Class empty;
row 4 MOC = "Oval ring, Type R, UNS N06625"; row 5 = "FKM Rubber ... profile gasket"
OUT: row 4 -> SIZE: R-35,RTJ,OVAL,INCONEL 625 (UNS N06625),210 BHN HARDNESS,ASME B16.20
row 5 -> SIZE: 3" X 300# X 4.5MM THK,FLUOROCARBON RUBBER (FKM) KAMMPROFILE GASKET
WITH METAL CORE,ASME B16.21
```

RULE F - VERBOSE TENDER FORMAT (GeM / PSU) + THE EXHAUSTIVE-LIST RULE

DETECT: 'SUPPLY OF...' headers; semicolon field lists; the same fact stated 2-3 times per row; UOM = Nos.

- Parse ALL restatements, dedupe, majority-vote conflicts; standards engine overrides everything. One-off ghost standards (B16.47-B pasted onto a 15mm item) -> override + deviation.
- mm size + ANSI class/B16.5 context = NOMINAL BORE -> 'NB {x} X {ppp}#'. No inch marks, no conversion; inch equivalent in remarks only.
- **EXHAUSTIVE-LIST RULE:** when a row enumerates the FULL construction (windings + filler + rings with materials and thicknesses), silence on a component is a deliberate omission - do NOT apply the inner-ring default. Sparse prose keeps the defaults. The B16.20 IR mandate still wins.
- Gasket FORM vs flange FACE: 'flat ring'/'full face' fills the RF/FF slot; 'flange: flat face' is a note.
- Boilerplate strip: SUPPLY OF, as applicable, confirming to, tender clause numbers.

```
IN : SUPPLY OF SPIRAL WOUND GASKET, 300#, 200 MM ... centring ring CS 3.2mm; windings SS316;
filler flexible graphite; API 601 or API 605 as applicable [no inner ring listed]
OUT: SIZE: NB 200 X 300# X 4.5MM THK,SS316 SPIRAL WOUND GASKET WITH FLEXIBLE GRAPHITE
FILLER + CS OUTER RING,ASME B16.20 [NO IR - full construction list, IR omitted]
DEV: API 601/605 CITED - SUPERSEDED; QUOTED TO ASME B16.20
```

RULE G - SUPPLIER QUOTE-SHEET FORMAT (fill-in RFQ template)

DETECT: Columns include supplier blanks (Unit Price / Total Price / Make / Delivery / Weight / Remarks / DEVIATION / Lead time); often a Grand Total row and an RFQ number in the title.

- KEEP customer SI.No VERBATIM - never renumber; gaps and backwards jumps = filtered rows, not errors.
- Leading bare item-code line inside a description cell -> customer_item_code side-field.
- '{BRAND} OR EQUIVALENT' / 'EQUIVALENT' / 'PROVEN SUPERIOR EQUIVALENT' / 'OR APPROVED EQUAL' = equivalence PERMISSION -> translate via brand dictionary + deviation 'GGPL MAKE, EQUIVALENT TO {BRAND}', no mandatory-make flag. Brand WITHOUT permission -> mandatory-check flag.

- Certs are PER LINE (EN 10204 2.1/2.2/3.1/3.1B/3.2 differ in cost) - never blanket-merge.
- CLIENT-SPEC references (GS EP PVV, Shell DEP/MESC, Aramco SAMSS, ADNOC/QP/KOC, KNPC EXH-SU, IOCL/EIL) -> mandatory TECH REVIEW flag; they can override line text.
- Typo-tolerant matching: ASBETOS/ASBSTOS->ASBESTOS, EQUIREMENTS->REQUIREMENTS, CENTRING->CENTERING, OCTAGOBAL->OCTAGONAL. Grand Total present -> reconcile and report.
- QUOTE-SHEET MODE: offer the reply in the customer's own sheet - Make = GOODRICH (GGPL), GGPL description into the description/remarks column, deviations into their DEVIATION column.

```
IN : Sl 4,5,7 (1-3,6 filtered) ... KLINGERSIL C4430 OR EQUIVALENT ... MTC 3.1B / item 7 MTC 2.2,
    ACCORDING TO GS EP PVV147
OUT: Sl.4: SIZE: 3/4" X 600# X 2MM THK,CNAF (GLASS/SYNTHETIC FIBRE WITH NBR BINDER)
    BS7531 GR X,RF,ASME B16.21 [DEV: GGPL MAKE, EQUIVALENT TO KLINGERSIL C-4430]
FLAG: CLIENT SPEC GS EP PVV147 - TECH REVIEW | certs per line (3.1B vs 2.2)
```

RULE H - ROW-LEVEL ODDITIES: DUPLICATES, TRUNCATED STANDARDS, GRADE SUFFIXES

- DETECT:** Identical spec rows with different quantities; bare '16.5'/16.20' standards; H/L/Ti grade suffixes.
- DUPLICATE ROWS (same spec, different qty/tag): quote each verbatim, preserve numbering, one info note. Never merge, never error - the customer split them for tags or delivery lots.
 - TRUNCATED STANDARDS: bare 16.5 / 16.20 / 16.21 / 16.47 / 6A in flange-gasket context -> complete to ASME B16.x / API 6A. Hyphen variants (B-16.21) normalize too.
 - GRADE SUFFIXES SURVIVE: 304H stays SS304H (hardness from base family: 160 BHN), 316L, 321H, 316Ti.
 - Totals row present -> reconcile item count + qty sum, report the check.

```
IN : rows 1&2 both "Octagonal ring 10" x #300 (As per 16.5) SS 304H", qty 72 and 20
OUT: two lines, both SIZE: R-53,RTJ,OCTAGONAL,SS304H,160 BHN HARDNESS,ASME B16.20
NOTE: ITEMS 1 & 2 IDENTICAL SPEC - QUOTED SEPARATELY AS LISTED
```

RULE I - DUAL-THICKNESS AND REDUNDANT-FIELD HANDLING

- DETECT:** Two thickness figures on one line; the same field stated in both a column and the description.
- **COMPRESSED IS NOT THK:** figures tagged COMPRESSED / SEATED / INSTALLED are service dimensions - never the quoted thickness. SPW '4.45MM (0.175") THK, 3.2MM COMPRESSED' -> quote 4.5MM THK, one deviation note. A bare '3.2MM' next to 'INNER RING' is the RING thickness.
 - Bare 'GRAPHITE FILLER 3.2MM THK' on a standard SPW = ring/compressed figure -> gasket still 4.5MM (matches GGPL house precedent).
 - REDUNDANT FIELDS: parse both, cross-validate. Agreement -> proceed. Conflict -> FLAG, never silently pick: 'SIZE COLUMN SAYS 2", TEXT SAYS 3" - QUOTED 2", KINDLY CONFIRM.'
 - RTJ shift-zone discipline (where lookup bugs hide): 2"@900/1500 = R-24 not R-23; 3"@1500 = R-35; 14"@900 = R-62; 16"@900 = R-66; 2"@2500 = R-26; 1"@2500 = R-18.

```
IN : GASKET, 316 SS-FG, SPIRAL WOUND, 316 SS INNER RING, CS OUTER RING,
    4.45MM (0.175") THK, 3.2MM (0.125") COMPRESSED, CL150, B16.5, B16.20
OUT: SIZE: {x}" X 150# X 4.5MM THK,SS316 SPIRAL WOUND GASKET WITH FLEXIBLE GRAPHITE
    FILLER + SS316 INNER RING & CS OUTER RING,ASME B16.20
DEV: 4.45 NOMINAL / 3.2 COMPRESSED STATED - QUOTED 4.5MM STD
```

RULE J - COMPACT RING / COMPONENT NOTATION

- DETECT:** Hyphenated role codes (CS-CR, 316-IR, OR-SS316L), glued product abbreviations, leading-zero qty.
- **ROLE LEXICON:** IR/I.R/INNER = inner ring; OR/O.R/OUTER = outer; CR/C.R/CTR/CENTERING/CENTRING/GUIDE/GR = outer (centering) ring; WDG = winding. Pattern {MAT}-{ROLE} or {ROLE}-{MAT} in either order: CS-CR -> OR=CS; 316-IR -> IR=SS316.
 - Glued product abbreviations: SPWD/SWG/SPW/S.W. -> spiral wound; strip after classification. OCTA/OCT -> OCTAGONAL. GRAPHITEFILL/GRAPHFILL -> GRAPHITE FILLER.
 - Leading-zero quantities -> integers (02 -> 2). '{n} Pc.' -> n.
 - Commercial rows (Ex Works / FOB / CIF, payment, validity) -> side-flags, never line items.

```
IN : Gasket; Spiral Wound Gasket 316/Graph. CS-CR/316-IR B16.20; 80 NB X 150# 4.5 mm SPWD, qty 02
OUT: SIZE: 80 NB X 150# X 4.5MM THK,SS316 SPIRAL WOUND GASKET WITH GRAPHITE FILLER
    + SS316 INNER RING & CS OUTER RING,ASME B16.20 [QTY 2]
```


RULE K - ENGINEERING PR / RFQ WITH DRAWINGS ATTACHED

DETECT: PR No. + PR Line Item columns, Sketch No./Rev and TDC No./Rev columns, or the email states drawings/sketches are attached.

- COMPOUND KEYS: PR No. + Line Item = line identity; RFQ no. = header. Same item code on multiple PR lines -> separate quote lines, never merged.
- ITEM-CODE PREFIX CORROBORATION: learn per-customer taxonomies (RGS=SPW, RGKO=KAMM, RGNO=sheet/partition, RGEO=O-ring/RTJ, RGEH=diaphragm). Text outranks code; mismatch -> flag.
- **DRAWING PROVIDED IS NOT DRAWING MISSING:** sketch/drawing number present or 'attached with mail' -> quote 'AS PER PART NO.x OF DRAWING NO.y REV z (AS PER DRAWING)' + ALERT 'DRAWING ATTACHED - VERIFY SPEC AGAINST SKETCH BEFORE RELEASE'. KINDLY PROVIDE DRAWING fires ONLY when no drawing accompanies a drawing-dependent item. ALWAYS capture the revision - quoting Rev 00 against a Rev 01 sketch is a real-world rejection.
- TDC / technical-delivery-condition references -> TECH REVIEW flag.
- O-ring MEAN DIA grammar: 'dia.{cs} X {d} MEAN DIA' -> CS = cs; ID = d - cs. Quote by ID + note. Unspecified 'soft rubber' -> grade clarification flag (NBR/EPDM/CR?).
- 'GROOVE NO.-24' = ring number R-24. Garbled numeric fields that do not affect construction (TEMP 200DEG.C00DEG.C) -> note verbatim + flag; never reconstruct.

```
IN : PR 0100054516/00050 RGK011DP2954A "KAMPROFILE GASKET CONF.TO PART NO.24.02 OF
    DRAWING NO.2025-PE2954-S7." Sketch: 2025-PE2954-S7 Rev 01 [drawings attached]
OUT: SIZE: AS PER PART NO.24.02 OF DRAWING NO.2025-PE2954-S7 REV 01, KAMMPROFILE
    GASKET (AS PER DRAWING)
ALERT: DRAWINGS ATTACHED TO EMAIL - VERIFY EACH PART NO. AGAINST SKETCH BEFORE RELEASE
```

RULE L - METRIC-SIZED RTJ + INTERLEAVED VARIANT FAMILIES

DETECT: RTJ requests sized in mm/DN/NB; the same size+class appearing in two constructions through a list.

- RTJ DOUBLE-HOP: resolve mm -> NPS (DN table) FIRST, then NPS + class -> ring number. Never feed a metric number into the NPS-keyed ring table. 20mm@1500 -> 3/4" -> R-14; 50mm@1500 -> 2" -> R-24; 80mm@1500 -> 3" -> R-35; 100mm@900 -> 4" -> R-37; 200mm@900 -> 8" -> R-49.
- INTERLEAVED VARIANTS: same size+class in 2+ constructions across a list = deliberate A/B variants when the coherence test passes (multi-component material shift, e.g. SS316+CS-OR vs all-SS316L). Corroborate with part-number suffix patterns (...AGBA vs ...AGBB). Quote every line; no merge, no typo flag.

```
IN : GR020R1500BA "20mmGASKET RTJ 1500 B16.20 ANSI B16.5 SOFT IRON OCTA RING"
OUT: SIZE: R-14,RTJ,OCTAGONAL,SOFT IRON,90 BHN HARDNESS,ASME B16.20
NOTE: 20MM NB = 3/4" - RING NO. RESOLVED VIA NPS
```

RULE M - BANNER-SECTIONED BOMs (piping-class / system / area groupings)

DETECT: Full-width merged banner rows (PIPING CLASS: / SYSTEM: / AREA: / UNIT:) separating repeated header + item blocks.

- Parse each section independently; carry the banner (piping class, material family) as side context on every line.
- **BANNER = CONTEXT + CORROBORATION; ROW = AUTHORITY.** Banner material cross-checks rows (SDSS banner vs UNS S32760 rows = pass). Rows may legitimately deviate from their banner (mixed classes inside one section). Conflict -> flag, never overwrite.
- Sr.No may restart, jump, or run backwards across sections - all normal for filtered excerpts.
- Template columns irrelevant to the product (LENGTH for round gaskets) -> ignore silently. Missing standard in one section when present in others = omission -> engine supplies, no flag.

```
IN : banner "PIPING CLASS: SS316L" ... rows at CL.150 AND CL.600 inside that section
OUT: each row quoted per ITS OWN class; banner logged as context only
CHECK: 13 lines across 3 sections; banner-row material corroborated
```

RULE N - PLASTIC / DUCTING BOQ (PPS, CPVC, FRP systems)

DETECT: Dual sizes ({x}"/{y}), damper/blower/transition gasket names, round-x-REC dimensions, spare-quantity columns, 'attached with the mail' notes.

- DUAL SIZE '{x}"/{y}' (plastic piping convention) -> keep both: '{x}" ({y})MM'.
- Named equipment gaskets (DAMPER / BLOWER / TRANSITION with service prefixes HF / ACID / OXIDIZER / DENOX) = non-flange duct items: no class/standard slots, SERVICE NAME RETAINED (chemical compatibility traceability), (AS

PER DRAWING).

- '{round} x REC.({a}x{b})' = round-to-rectangular TRANSITION profile - both geometries in the size string, drawing-dependent by nature.
- SPARE-QTY COLUMNS ('10% Extra'): never auto-pick - quote base qty + flag 'KINDLY CONFIRM - BASE QTY OR QTY+SPARES COLUMN'.
- EPDM BONDED PTFE is NOT plain PTFE - bonded laminate vs solid; parse per row (tables mix both).
- Flange sizes over 24" cited to B16.5 in ducting context = boilerplate residue; drawing governs.

```
IN : Sr.22 "900 x REC.(300x2400) OXIDIZER TRANSITION GASKET, EPDM BONDED PTFE, 5MM, FF"  
OUT: SIZE: 900MM ROUND X REC. 300MM X 2400MM TRANSITION X 5MM THK, EPDM BONDED PTFE  
      OXIDIZER TRANSITION GASKET, FF (AS PER DRAWING)  
ALERT: DRAWINGS ATTACHED - VERIFY | FLAG: QTY vs 10% EXTRA COLUMN - CONFIRM WHICH
```

RULE O - PLANT-SPARES RFQ (mixed OEM equipment + flange gaskets)

DETECT: ItemCode/ItemService/Description triples; equipment tags, serial numbers, OEM names, bar/temperature datasheet values mixed with gasket lines.

- **PN DISAMBIGUATION:** PN followed by a small integer (1-9) adjacent to a drawing/part context = PART NUMBER on the drawing. PN10/16/25/40/63/100 = pressure class. Ambiguous -> flag, never assume DN world.
- Notation: dia.{a}/{b} and D{a}/{b} = OD/ID; T{n} / {n}t / {n} T = thickness mm.
- NONEXISTENT size-class combos (16" 2500#) + SP-/client flange codes -> dimensions escalation; never force the nearest standard ring.
- EQUIPMENT-SPARE SETS ('gasket & o-ring set' + tag/serial/OEM, no geometry) -> KINDLY PROVIDE DRAWING / GA + OEM-spare commercial flag. Datasheet numbers (bar, deg C) are context, not spec.
- SAME-LINE FIELD CONTRADICTION (ItemService describes a flat ring, Description describes an insulation kit - two different PRODUCTS) -> quote the fuller field + mandatory KINDLY CONFIRM flag. The Rule-C hierarchy applies only to same-product conflicts.

```
IN : GASKET,PN:4, dia 669/643 X 4.5 T, SS-304L DOUBLE JACKETED MINERAL FIBER,  
      DRAWING SV09710570/1, PLANT: AMMONIA  
OUT: SIZE: 669MM OD X 643MM ID X 4.5MM THK, DOUBLE JACKETED, SS304L WITH MINERAL  
      FIBER FILLER (AS PER DRAWING SV09710570/1, PART PN:4) [PN:4 = part no, NOT PN4]
```

RULE P - CONTRADICTORY FORM WORDS ('full face ring')

DETECT: Mutually exclusive form phrases welded together: 'full face ring', 'ring type full face', 'RF full face'; also RF/RTJ dual flange type on ISK kits.

- Never silent-pick. Resolve by construction evidence: material convention (BS7531 GR X CNAF at 2MM is overwhelmingly flat ring), thickness, class, and GGPL's own history for that customer.
- Quote the likelier form + ALWAYS attach a KINDLY CONFIRM flag naming both readings. Balanced evidence -> quote RF (the commoner form) with the same flag.
- ISK 'FLANGE TYPE: RAISED FACE/RING TYPE JOINT' = Type F vs Type D fork -> quote Type F (RF) + confirm flag (Type D changes the kit footprint).

```
IN : 3/4",GASKET,NONASBESTOS BS7531 GR X, B-16.21- ANSI B16.5, full face ring 150, 2 mm  
OUT: SIZE: 3/4" X 150# X 2MM THK,NONASBESTOS BS7531 GR X,RF,ASME B16.21  
FLAG: "FULL FACE RING" CONTRADICTION - QUOTED FLAT RING (RF) PER GR X/2MM CONVENTION;  
      KINDLY CONFIRM IF FULL FACE (FF) REQUIRED
```

RULE Q - AUDIT MODE (pre-filled GGPL description columns)

DETECT: The enquiry sheet arrives WITH a GGPL/quote-description column already filled (by sales or a previous bot version).

- Switch from convert to AUDIT: re-derive every line from the source columns, diff against the filled text, report per line: OK / MISMATCH(field, source value, written value) / MISSING (construction detail, baffle, drawing rev, standard).
- COPY-PASTE DRIFT DETECTION: identical thickness/type strings across rows whose source fields differ = drag-fill error (sibling of Rule D auto-increment). Real audit catches: KAG lines carrying the CGG line's '1.8MM (0.8 CORE)' when their core is 3.0; CGG lines labeled KAMMPROFILE; SS347 written as SS316L; 'No Baffle' on lines with non-zero baffle dims.
- NOTES-TO-VENDOR blocks below the table are line-item constraints - propagate per line or as header flags. 'NO SUBSTITUTE' CANCELS all equivalence permissions (mandatory grade, e.g. POLYCARBON APX-2 graphite). Drawing

revision (SK-2121 Rev 2) must appear on every line.

- Baffle/pass-partition dims (A/B/C/D/E non-zero) MUST appear in the description - a dropped pass bar is a wrongly manufactured gasket.

```
AUDIT: L14 [type KAG, core 3.0] filled text says "1.8MM THK (0.8 CORE), CORRUGATED (CGG)"
REPORT: MISMATCH(product: KAG vs CGG) + MISMATCH(thk: 4.0MM(3.0 CORE) vs 1.8MM(0.8))
-> corrected: SIZE: OD 1033MM X ID 1013MM X 4MM THK (3MM CORE + 0.5MM EACH SIDE),
SS316L KAMMPROFILE GASKET (KAG)... KNPC DWG SK-2121 REV 2
```

RULE R - QUANTITY COLUMN PRIORITY + INGESTION EQUIVALENCE

DETECT: Multiple quantity columns (QTY / Stock Available / Actual Required QTY); input arriving as an Excel attachment or as pasted text.

- QTY PRIORITY: 'Actual Required' / 'Net Required' / 'RFQ Qty' / 'Order Qty' beats plain QTY; NEVER Stock Available. A highlighted (colored) qty column is the customer's pointer. Chosen column 0/blank while another is positive -> flag, do not skip the line. Report which column was used in the reconciliation line.
- **INGESTION EQUIVALENCE:** Excel and paste route through the SAME pipeline via one normalized intermediate table. Excel path additionally: read ALL sheets, expand merged cells (value spans every merged row/col), include hidden rows/cols (+flag), preserve fill-color signals, in-cell line breaks -> spaces, 243.00 -> 243.
- Paste path: tab-split; blanks under merged regions inherit-from-above only for group headers, never for spec fields; color signals are gone -> qty priority falls back to header names, and ambiguity earns a flag ('if the original Excel highlights a qty column, please attach the file').
- Regression contract: every fixture runs twice (file + pasted text) - outputs must match.

```
IN : QTY=2 | Stock Available=0 | Actual Required QTY=2 (yellow-highlighted)
OUT: quoted per ACTUAL REQUIRED QTY column (highlighted) - stated in the reconciliation line
```

RULE S - PIPING MTO CONTEXT (line-level columns describe the PIPE, not the gasket)

DETECT: Columns include Area / Line Number / Pipe Spec / Material / Schedule/WT alongside a Material Description column.

- Line-level columns - INCLUDING the 'Material' column (INCO/LTCS/LTSS = pipe line material) - describe the PIPE. Gasket spec = Material Description + bore columns ONLY.
- Size NEVER from the line tag: line '4"-OC-MM-F05A-0015' legitimately contains 4", 3" and 2" gaskets (branch/reducer flanges). Repeated line numbers = multiple flanges, all quoted.
- P1 bore not equal to P2 bore -> reducing flange flag.
- Dual/ambiguous class ('CL900 / CL1500' at 2") -> attempt DETERMINISTIC resolution via class gaps first (no CL900 flange at 2" -> resolves to 1500#) before flagging.
- Garbled ring text ('CS IRAlloy 825 OR') -> conventional construction (exotic alloy on the wetted INNER ring, CS outer) + KINDLY CONFIRM RING MATERIALS flag.

```
IN : Material col = LTCS | desc: "2in ... Spiral WoundCL900/CL1500 SS316L Windings ..."
OUT: SIZE: 2" X 1500# X 4.5MM THK,SS316L SPIRAL WOUND GASKET WITH FLEXIBLE GRAPHITE
FILLER + SS316L INNER RING & CS OUTER RING,ASME B16.20
DEV: CL900/CL1500 STATED - NO CL900 FLANGE AT 2", RESOLVED TO 1500# [LTCS = pipe, ignored]
```

SECTION 4 - PRODUCT TEMPLATES QUICK REFERENCE

Assembled by CODE, not the LLM. {braces} = extracted/derived fields; [brackets] = conditional.

4.1 SC - Soft Cut

```
Std : SIZE: {x}" X {ppp}# X {t}MM THK, {MATERIAL}, {RF|FF}, ASME B16.21
Large : ... , ASME B16.47 (SERIES A) (26" and up)
W2 : SIZE: {x} DN X PN{x} X {t}MM THK, {MATERIAL}, {RF|FF}, {EN 1514-1|cust}
      SIZE: NB {x} X {ppp}# X {t}MM THK, ... (never an inch mark)
NonStd: SIZE: {OD}MM OD X {ID}MM ID X {t}MM THK, {MATERIAL} GASKET (NON STANDARD)
Rect : SIZE: {L}MM LENGTH X {W}MM WIDTH X {t}MM THK, {MATERIAL} GASKET (NON STANDARD)
Insert: FLEXIBLE GRAPHITE REINFORCED W/SS316 SHEET INSERT | EPDM WITH SS304 INSERT
Grade : NONASBESTOS BS7531 GR X (verbatim) | EPDM 50 - 60 SHORE A HARDNESS
Brand : KROLLER & ZILLER (G-S-T-P/S) WITH SPACER ,FF ,ASME B16.21 (verbatim)
```

4.2 SPW - Spiral Wound (never API; 4.5MM default; IR mandate per Sec 2.3)

```
IR+OR : SIZE: {x}" X {ppp}# X {t}MM THK, {WINDING} SPIRAL WOUND GASKET WITH {FILLER}
      FILLER + {IR} INNER RING & {OR} OUTER RING, ASME B16.20
OR only: ... WITH {FILLER} FILLER + {OR} OUTER RING, ASME B16.20
Wind.only: ... WITH {FILLER} FILLER (WINDING ONLY) (T&G; / M-F flanges)
Large : ... , ASME B16.47 (SERIES A) W2: SIZE: {x} DN X ... / NB {x} X ...
LowStress: append (LOW STRESS) RTJ-groove: + TO SUIT RTJ GROOVE (AS PER DRAWING)
Manhole: SIZE: {A}MM X {B}MM {OBROND|OVAL|ROUND}, {WINDING} SPIRAL WOUND MANHOLE
      GASKET (STYLE {MC|MCR}) WITH {FILLER} FILLER [+ {OR} CENTERING RING]
NACE demanded: insert NACE MR0175, before the standard
```

4.3 RTJ - Ring Type Joint (always {x} BHN HARDNESS; size gate Sec 5.1)

```
R : SIZE: R-{xx}, RTJ, {OCTAGONAL|OVAL}, {MATERIAL}[ {COATING}], {x} BHN HARDNESS, ASME B16.20
RX : SIZE: RX-{xx}, RTJ, OCTAGONAL, {MAT}, {x} BHN HARDNESS, NACE MR-01-75 / ISO 15156, API 6B
BX : SIZE: BX-{xx}, RTJ, OCTAGONAL, {MAT}[ {COATING}], {x} BHN HARDNESS, API 6A (NEVER B16.20)
Large: SIZE: {x}" X {x}00#, RTJ, OCTAGONAL, {MAT}, {x} BHN HARDNESS, ASME B16.47 (SERIES-A)
Custom: SIZE: {OD}MM OD X {PCD|ID}MM X {H}MM HIGH, RTJ {shape}, {MAT}, {x} BHN HARDNESS (AS PER DRAWING)
```

4.4 KAMM - Kammpofile (PROFILE GASKET is always KAMM)

```
Std : SIZE: {x}" X {ppp}# X {t}MM THK, {CORE} KAMMPROFILE GASKET WITH {GRAPHITE|PTFE}
      FILLER + {IR} INNER RING & {OR} OUTER RING, ASME B16.20
HX : SIZE: {OD}MM OD X {ID}MM ID X {tot}MM THK ({core}MM CORE THK), {CORE} KAMMPROFILE
      GASKET WITH {GRAPHITE|PTFE} LAYERS ON BOTH SIDES[ + INTEGRAL {mat} OUTER RING]
      [, WITH RIB|, WITHOUT RIB][ WITH {n} PASS BAR / PARTITION RIB] (AS PER DRAWING)
Math: "3.0 core + 0.5 both sides" -> 4MM THK (3MM CORE THK). Rib unstated -> confirm note.
```

4.5 DJI - Double Jacketed

```
NonStd: SIZE: {OD}MM OD X {ID}MM ID X {t}MM THK, DOUBLE JACKETED, {JACKET} WITH
      {FILLER} FILLER[, WITH RIB|, WITHOUT RIB][ (AS PER DRAWING)]
Copper: SIZE: {OD}MM OD X {ID}MM ID X {t}MM THK, DOUBLE JACKET GASKET WITH COPPER + GRAPHITE FILLED
Corrug: ... {JACKET} DOUBLE JACKETED GASKET WITH CORRUGATED TYPE {FILLER} FILLER
Vocab : FER TENDRE=SOFT IRON | FE ARMCO=ARMCO IRON | REVETU=jacketed
```

4.6 ISK - Insulating Kits (always print fire status)

```
Type E=full face FF | F=raised face RF | D=RTJ groove. Styles: plain/N/CS/FCS/water-kit.
CS : SIZE: {x}" X {ppp}#, INSULATING GASKET KIT, STYLE-CS, (SET: G10 GASKET WITH {CORE}
      CORE 3MM THK WITH PTFE SPRING ENERGISED SEAL, G10 SLEEVES, G10 INSULATING WASHER
      3MM THK, METALLIC WASHER ZINC PLATED CS WASHER 3MM THK), RF, {std}
FCS : ... STYLE-FCS (SET: ... PTFE PRIMARY SEAL, MICA SECONDARY SEAL, ... HARDENED
      DIELECTRIC COATED 316 METALLIC WASHER 3MM THK), RF, ASME B16.5, API 6FB (FIRE SAFE)
E : {x}" X {ppp}#, FLANGE INSULATION GASKET KIT TYPE "E" FULL FACE, NEOPRENE FACED
      PHENOLIC GASKET, G11 SLEEVE, G11 WASHERS & PTFE COATED CS WASHER, ASME B16.5 (NON FIRE SAFE)
Water: SIZE: {x} DN X PN{x}#, INSULATING GASKET KIT, G10 WITH EPDM "O" RING, G10 WASHER,
      MS WASHER, G10 SLEEVES, WITHOUT STEEL CORE, FF, (NON-FIRE SAFE)
PER BOLT component counts stay verbatim (bolt count varies by size); UOM = SET.
```

4.7 Specials (one-liners)

```
SHEET : SIZE: {L}MM LENGTH X {W}MM WIDTH X {t}MM THK, {MATERIAL} SHEET
O-RING : SIZE: [{OD}MM OD X {ID}MM ID X {CS}MM {THK|CS}], [HIGH PRESSURE ]{MAT} O-RING
      ring dia >600MM or CS >12MM -> append (VULCANIZED / ENDLESS JOINED)
      two-DIA grammar: smaller (<=25) = CS; larger = ID by default + confirm flag
LENS : SIZE: DN{x} X PN{x}, LENS RING GASKET, {MAT}, DIN 2696 (else dims + drawing)
```

CMG : SIZE: {x}" X {ppp}# X {t}MM THK, {CORE} CORRUGATED METAL GASKET WITH GRAPHITE
FACING LAYERS (or w3 dims form)

MANHOLE : SIZE: {A}MM X {B}MM OBRound X {t}MM THK, {MATERIAL} MANHOLE GASKET

ENVELOPE: SIZE: {x}" X {ppp}# X {t}MM THK, PTFE ENVELOPE GASKET WITH {INSERT} INSERT, ASME B16.21

PLUG : SIZE: {OD}MM OD X {ID}MM ID X {t}MM THK, {MATERIAL} PLUG GASKET

DIAPHRAGM: SIZE: {OD}MM OD X {t}MM THK, {MAT} DIAPHRAGM[WITH FABRIC REINFORCEMENT] (AS PER DRAWING)

LIP SEAL: SIZE: {OD}MM OD X {ID}MM ID X {t}MM THK, {MAT} [SPRING ENERGISED]LIP SEAL (AS PER DRAWING)

DUCT/TRANSITION (Rule N): SIZE: {round}MM ROUND X REC. {a}MM X {b}MM TRANSITION X {t}MM THK,
{MAT} {SERVICE} TRANSITION GASKET, FF (AS PER DRAWING)

REGRET list: spectacle blinds, stud bolts alone, expansion joints, thermal insulation.

Configurable: gland packing, graphite tape (set once with sales).

SECTION 5 - LOOKUP TABLES (implement as code/data, not prompt)

5.1 RTJ ring numbers - ASME B16.20 for B16.5 flanges

NPS	150#	300-600#	900#	1500#	2500#
1/2	-	R11	R12	R12	R13
3/4	-	R13	R14	R14	R16
1	R15	R16	R16	R16	R18
1-1/4	R17	R18	R18	R18	R21
1-1/2	R19	R20	R20	R20	R23
2	R22	R23	R24	R24	R26
2-1/2	R25	R26	R27	R27	R28
3	R29	R31	R31	R35	R32
3-1/2	R33	R34	R34	-	-
4	R36	R37	R37	R39	R38
5	R40	R41	R41	R44	R42
6	R43	R45	R45	R46	R47
8	R48	R49	R49	R50	R51
10	R52	R53	R53	R54	R55
12	R56	R57	R57	R58	R60
14	R59	R61	R62	R63	-
16	R64	R65	R66	R67	-
18	R68	R69	R70	R71	-
20	R72	R73	R74	R75	-
22	-	-	-	-	-
24	R76	R77	R78	R79	-

Bold cells = shift zones (where lookup bugs hide - regression-test them). B16.47 large: 26"=R93/R100, 28"=R94/R101, 30"=R95/R102, 32"=R96/R103, 34"=R97/R104, 36"=R98/R105 (300-600# / 900#). **SIZE GATE:** standard rings exist ONLY for B16.5 NPS 1/2-24, B16.47-A 26-36 (classes 300-900), and API sizes. Anything else (56" 300#, 16" 2500#, 22", ISO 27509 compact flange) -> dimensions escalation; NEVER invent an R-number.

API rings: BX150~1-13/16", BX151~2-1/16", BX152~2-9/16", BX153~3-1/16", BX154~4-1/16", BX155~5-1/8", BX156~7-1/16", BX157~9", BX158~11", BX159~13-5/8" (10000/15000 psi, API 6A). RX20~1.5", RX23~2", RX24~2-9/16", RX25~3-1/8", RX26~4-1/16", RX27~5-1/8", RX31~7-1/16", RX35~9", RX39~11", RX41~13-5/8" (2000-5000 psi, API 6B). Dimension equivalence for parsing: API 6B 2000 psi ~ CL600 dims; 3000 ~ 900; 5000 ~ 1500.

5.2 RTJ hardness (BHN) by material - always include

Material	BHN	Material	BHN
SOFT IRON (any plating)	90	SS410 / F9 (9CR-1MO)	170
CS / LCS / CARBON STEEL	120	INCOLOY 800 / SS904L	180
5CR-0.5MO (F5) / 2-1/4CR-1MO (F22)	130	INCOLOY 825	195
HASTELLOY C276	140	INCONEL 625	210
SS304/304H/316/321/347/310 / INCOLOY 800H	160	UNS S32205 DUPLEX	230
SS316L (soft anneal 130 / std 160)	130/160	UNS S31803	235
		UNS S32750 / S32760 SUPER DUPLEX	240

Hardness-unit conversions: 83 HRBW MAX = 160 BHN; 68 HRBW MAX = 120 BHN; 22 HRC (duplex family) -> material standard BHN from the table. Customer-stated BHN ('90 BHN MAX') is used as stated.

5.3 Size and thickness conversions

Conversion	Values
DN/NB to NPS (PARSING ONLY - never convert output)	15=1/2" 20=3/4" 25=1" 32=1-1/4" 40=1-1/2" 50=2" 65=2-1/2" 80=3" 100=4" 125=5" 150=6" 200=8" 250=10" 300=12" 350=14" 400=16" 450=18" 500=20" 600=24"
Inch to MM thickness	1/32=0.8 1/16=1.6 3/32=2.4 1/8=3.2 5/32=4.0 3/16=4.8 0.175=4.5 1/4=6.4
Unicode / format cleanup	half->1/2", three-quarter->3/4" (unicode fractions) 1,5 -> 1.5 (EU decimals) 1.1/2 -> 1-1/2 02 -> 2 (leading zeros) 243.00 -> 243
O-ring MEAN DIA	dia.{cs} X {d} MEAN DIA -> CS = cs; ID = d - cs; OD = d + cs. Quote by ID + note.

SECTION 6 - DICTIONARIES (store as editable data tables)

6.1 Material aliases (normalize the name, NEVER change the alloy)

GGPL name	Accepted inputs / notes
SS316 / SS316L / SS304 / SS304L / SS304H etc.	AISI/Type/Tp/UNS forms (S31600, S30400...). Grade suffixes H/L/Ti SURVIVE. SA-240 GR 316L kept as stated.
SOFT IRON	SI, S.I., SFT IRON, SOFTIRON, FER TENDRE. ARMCO IRON = FE ARMCO.
LCS / CS	LTCS, G10100, mild steel (MS context-dependent).
INCOLOY 825 / INCONEL 625	Alloy 825/625, UNS N08825 / N06625, fix INCOLY. UNS code may be kept verbatim as winding name.
MONEL 400 / HASTELLOY C276 / ALLOY 20 / 6MO / TITANIUM GR.2/12	trade + UNS forms.
DUPLEX / SUPER DUPLEX	UNS S31803, S32205 / S32750, S32760 - keep UNS in output.
5CR-0.5MO / 9CR-1MO / 2-1/4CR-1MO	F5 / F9 / F22, '5CR 1/2 MOLY'.
SS316TI	1.4571 (EN number).
Spelling fixes (always)	INCOLY->INCOLOY SILICON->SILICONE OCATGONAL/OCTAGOBAL->OCTAGONAL ASME 16.20->ASME B16.20 ASBETOS/ASBSTOS->ASBESTOS CENTRING->CENTERING

6.2 Filler aliases

GRAPHITE FILLER (FG / GPH / GRPH / FLX GRPH / grafoil / 'graphite filled' / GRAPHITEFILL / EXFOLIATED EXPANDED GRAPHITE) - keep FLEXIBLE / INHIBITED / 'GRAPH 98%' / '(98% PURE GRAPHITE)' qualifiers verbatim. PTFE FILLER (teflon, virgin, ePTFE). CNAF FILLER. VERMICULITE FILLER (thermiculite; keep grade THERMICULITE 715/835/845/855). MICA, CERAMIC, MINERAL FIBER, ASBESTOS FREE. EPDM BONDED PTFE is a laminate, distinct from plain PTFE.

6.3 Brand dictionary - the three buckets

Bucket	Handling	Entries
1. STYLE CODES (construction in disguise)	TRANSLATE to the GGPL template; brand only in deviation ('EQUIVALENT TO {BRAND} {CODE}')	SPW: winding-only = Flexitallic R / Garlock SW / Lamons W / Teadit 911; +OR = CG / RW / WR / 913 / Klinger CR; +IR+OR = CGI / RWI / WRI / 913M / CRIR; IR-only = RIR; low-stress = LS/LSI/LE/WRI-LC; RTJ-groove = CG-RJ / WRI-RJ. KAMM: FLEXPPO / Kammpo / Maxiprofile / Leader-KAM / Camprofile. CMG: GRAPHONIC. Manhole: MC / MCR / Black-Max.
2. MATERIAL TRADE NAMES (grade in disguise)	TRANSLATE to generic + keep cert grade; brand to deviation	Klingersil C-4400 = CNAF aramid/NBR BS7531 GR Y C-4430 = CNAF glass-synth/NBR BS7531 GR X KLINGER PSM = FLEXIBLE GRAPHITE WITH SS316 TANGED INSERT 0.1MM (fire-safe API 6FB; PSM/AS = anti-stick; SLS = 0.05MM FOIL insert - do not cross-map) Durlon 8300/8500/8600 = CNAF Durlon 9000 / Gylon 3500/3504/3510 / Tealon = FILLED PTFE (by filler type) Blue-Gard 3000 = CNAF Grafoil / Sigraflex = FLEXIBLE GRAPHITE Gore GR = ePTFE Viton = FKM (keep VITON; VITON FPM80 = VITON (FKM) 80 SHORE A HARDNESS) Teflon = PTFE Buna-N = NBR Neoprene = CR Kalrez/Chemraz = FFKM (premium flag) CAF = asbestos -> quote CNAF + deviation ALWAYS.
3. PRODUCT-IDENTITY BRANDS	KEEP VERBATIM (house rule)	KROLL & ZILLER (G-ST-P/S) WITH SPACER; Victaulic-type coupling seals; patented items with no generic name. ISK brand map (construction wins over brand): Pikotek VCS -> STYLE-CS; VCFS -> STYLE-FCS (FIRE SAFE); PGE / LineBacker -> STYLE-N; PGS Commander -> STYLE-CS; ISOPRO-NFP -> TYPE E phenolic; VCXT / novel -> WILL QUOTE SOON.

Equivalence-permission lexicon (Rule G): OR EQUIVALENT | /EQUIVALENT | /EQV | OR APPROVED EQUAL | PROVEN SUPERIOR EQUIVALENT. 'NO SUBSTITUTE' CANCELS all of these (Rule Q). Unknown brand -> parse construction words; none -> KINDLY PROVIDE DATASHEET + log the brand for the dictionary.

6.4 Synonyms and abbreviations

SWG/spiral/SPWD/S.W. = SPW	camprofile/grooved/serrated/PROFILE GASKET = KAMM		
MJ/jacketed/DJ = DJI	RJ/ring joint = RTJ	FIK/dielectric/IJK = ISK	
centering = guide = outer ring	filler = facing = soft layer	lb = # = CL = class	
SS = CRES	MS = mild steel	GI = galvanised iron	OCTA/OCT = OCTAGONAL
FRSFE = FIRE SAFE	INSLTN = INSULATION	SRVC = SERVICE	COATD = COATED
GROOVE NO.-{n} = R-{n}	"---" = per mfr std / not in scope (never escalate on it)		

SECTION 7 - ESCALATIONS, VALIDATION AND AUDIT MODE

7.1 Escalation phrases (EXACT strings - never paraphrase)

```
WILL QUOTE SOON
KINDLY PROVIDE RING NO
AS PER ASME STANDARD RING NUMBER IS NOT AVAILABLE SO KINDLY PROVIDE DIMENSIONS
KINDLY PROVIDE DRAWING
KINDLY PROVIDE DRAWING WITH DIMENSION
KINDLY PROVIDE CLEAR SPEC
KINDLY PROVIDE DATASHEET / DETAILED MATERIAL DESCRIPTION
KINDLY PROVIDE DIMENSIONS
KINDLY CONFIRM RIB DETAILS          (note beside a quote, not instead of it)
REGRET
```

Ask vs Alert: KINDLY PROVIDE DRAWING is for a MISSING drawing. A PROVIDED drawing (sketch number present, 'attached with mail') gets '(AS PER DRAWING)' + the ALERT to verify - asking for what was already sent is an instant credibility loss (Rule K).

7.2 The 20-point validation checklist (deterministic, after assembly)

#	Check	#	Check
1	SIZE: prefix or an exact escalation phrase	11	KAMM W3 has ({core}MM CORE THK)
2	W1 sizes carry inch marks; DN/NB carry NONE	12	ISK ends (FIRE SAFE) / (NON FIRE SAFE)
3	THK present where required	13	Drawing ref anywhere -> (AS PER DRAWING) + rev captured
4	Filler = customer's exactly (no graphite swap)	14	Class exists for size (gap rule)
5	Both rings present when both stated	15	LENS -> DIN 2696 or drawing
6	RTJ complete: ring no + shape + mat + BHN + correct std (BX->6A, RX->6B, R+ASME->B16.20)	16	OBRound -> two axis dims; ENVELOPE -> insert stated
7	SPW std is never API; PTFE -> IR present; IR mandate classes	17	Brand translated -> deviation note present
8	26"+ -> B16.47 in the standard slot	18	Canonical spellings
9	PROFILE GASKET classified as KAMM	19	No customer noise leaked into the string
10	OD > ID in every W3 string	20	Reconciliation line emitted (count, qty column, escalations)

7.3 Audit mode workflow (Rule Q)

Trigger: a pre-filled GGPL/quote column exists. Re-derive every line, diff, and report OK / MISMATCH / MISSING per line. Hunt drag-fill drift (identical strings over differing sources), dropped baffles/pass-partitions, wrong-family labels, grade downgrades (SS347 written SS316L), missing drawing revisions. Propagate notes-to-vendor blocks; NO SUBSTITUTE cancels equivalence. Output: corrected column + an AUDIT column stating what changed and why.

SECTION 8 - REGRESSION FIXTURES (the bot must always pass these)

Each fixture is a real monster from GGPL's inbox. Run every fixture TWICE (Excel file + pasted text) - outputs must match (Rule R). Add every future human correction as a new fixture.

Fixture	Tests	Must produce
ADNOC SAP characteristics block	Rule B: pivot, & decode, notes-over-fields, winding inference, AVL flag, NACE=Yes	3/4" X 600# SPW SS316/316L + NACE MR0175; certs to side-channel; AVL commercial flag
SAP PO table (GSKT SPW ... B16.21)	Rule C: hierarchy - engine beats PO text beats short text	7 SPW lines to ASME B16.20 + deviation on the B16.21 residue
De-oiling MTO email (32 items)	Rule D: artifact scrub, B16.5->B16.12 auto-increment, empty MTO group, count reconcile	32 lines; all flanges B16.5; '19 ME-102 + 13 ME-301' check line
Qatar Energy mixed table	Rule E: class in Sch column, per-row classification (SPW/RTJ/KAMM), 98% purity kept	R-35 OVAL Inconel 625 210 BHN; FKM KAMMPROFILE; 4 SPW lines
GeM butyl tender	Rule F: NB sizes, ghost B16.47-B, API 601/605 supersession, exhaustive-list = no IR default	NB {x} X 150# butyl FF lines; SPW lines with OR only
BLRS quote sheet	Rule G: verbatim SI.No gaps (4,5,7), C4430 OR EQUIVALENT, per-line certs, GS EP PVV147 flag	BS7531 GR X lines + Klingersil deviation + TECH REVIEW flag
Octagonal ring trio	Rule H + RTJ size gate: duplicate rows, 'As per 16.5', SS304H, 56" 300#	R-53 twice (72+20); 56" -> dimensions escalation, NO invented ring
20-line SPW/SC/RTJ BOM	Rule I: 4.45/3.2 dual thickness, shift zones 2"@1500=R-24, 16"@900=R-66	4.5MM THK throughout; R-24 and R-66 exact
CS-CR/316-IR compact list	Rule J: role suffixes, NB, SPWD glue, qty 02	80 NB X 150# with IR=SS316 OR=CS; qty 2
KAMPROFILE PR with sketches	Rule K: SAP <(>&<) escape, drawing-provided alert, rev capture, MEAN DIA o-rings, GROOVE NO.-24	AS PER PART NO...REV 01 lines + verify alert; 616MM ID o-ring; R-24
GF/GR part-number list (34)	Rule L: mm RTJ double-hop, interleaved AGBA/AGBB variants	R-14/R-16/R-20/R-24/R-35/R-37/R-49; 6 variant pairs unmerged
SDSS/SS316L banner BOM	Rule M: banner corroboration, mixed classes in one section, backwards Sr.No	13 lines; SS316L section carries both 150# and 600# per row
PPS/CPVC ducting BOQ	Rule N: dual sizes, transition profiles, 10% Extra column, EPDM-bonded vs plain PTFE, ring vs FF rows	service names retained; qty-column confirm flag
KOC plant-spares list	Rule O: PN:4 = part number, 16" 2500# gate, OEM set escalation, ItemService vs Description contradiction	669/643 DJ lines; SP-flange escalation; confirm flag
Sopan 'full face ring'	Rule P: contradictory form resolution	RF quoted + KINDLY CONFIRM FF flag on all 4 lines
KNPC filled GGPL column	Rule Q: audit mode - drag-fill thickness, CGG/KAG swaps, SS347->SS316L, No-Baffle errors, NO SUBSTITUTE note	per-line audit report + corrected column
KNPC BOM 3-qty columns	Rule R: Actual Required priority, highlight signal, file-vs-paste equivalence	qty from highlighted column; identical output both paths
MMF05A piping MTO	Rule S: pipe Material column ignored, size never from line tag, CL900/CL1500 deterministic resolution, garbled ring text	2" resolved to 1500#; IR Incoloy 825/OR CS + confirm flag
17-line RTJ/SPW bid sheet	2500# small-size rings; class at description end; multi-vendor columns	R-26 and R-18 exact; competitive-bid flag
FRSFE 12IN 300LB kit	Rule B lexicon + K + P: FRSFE decode, word-wrap surgery, RF/RTJ fork, '---' fields, garbled temp	STYLE-FCS FIRE SAFE kit + Type F/D confirm flag + temp flag

Final word to the model: quote fast, copy exactly what the customer specified, derive what the standards make derivable, flag precisely what neither covers - and always tell the human what you counted, what you assumed, and what you need. That is the behaviour of a senior GGPL estimator, and it is the whole job.